

AetherGrid

Peer Distributed Storage Marketplace & Zero-Trust Personal Cloud Backup

1. What's New Since Last Week

Since last week, the application has evolved from a foundational concept into an end-to-end operational **two-sided peer storage marketplace** with enterprise-grade zero-trust cryptography and dedicated hardware daemon support:

Zero-Knowledge Cryptographic Core

- Implemented **HKDF-SHA256** (RFC 5869) key derivation per customer and per object.
- AES-256-GCM authenticated encryption** with 128-bit authentication tags ensuring storage providers cannot read filenames or plaintext contents.

Dedicated Physical Node Daemon

- Engineered a production storage worker daemon for external/internal drives (configured for `D:\AetherGridStorage`).
- Automated chunking (2MB slices), heartbeat monitoring (30s intervals), and real-time offline failover.

Mobile Cloud Backup Simulator

- Built an interactive in-browser phone backup simulator (`/mobile-simulator`) modeling real iOS/Android cloud backups.
- One-tap photo roll, contacts, and personal vault backup with live chunk distribution progress.

Two-Sided Marketplace & Ledger

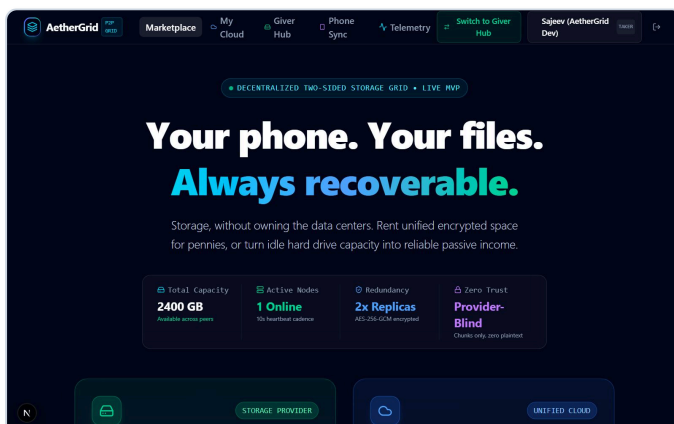
- Takers:** Automated 20GB tier allocation, file previews, and 60-second scoped signed download tokens.
- Givers:** Dynamic monetization engine tracking allocated GB-hours and monthly earnings in INR.

Security & Reliability Verification:

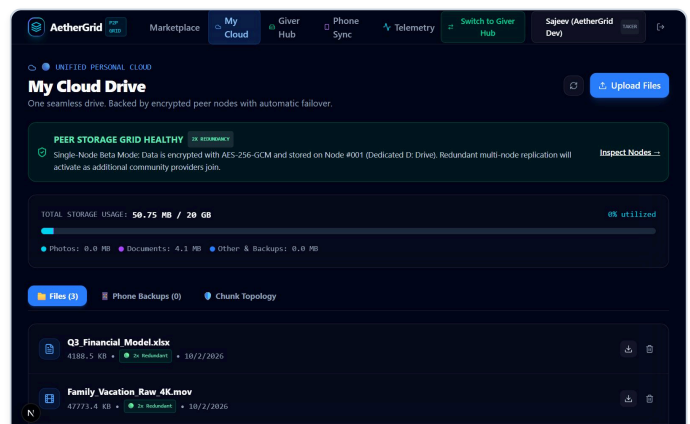
Completed a comprehensive 28-vector adversarial penetration audit (including NTFS alternate data stream protection, directory traversal escapes, token hijacking, and tamper detection) with a 100% pass rate (59/59 automated unit & integration tests green).

2. Current Version Screenshots (Captured from Localhost)

The following high-resolution screenshots reflect the live, working application running on `http://localhost:3000`:



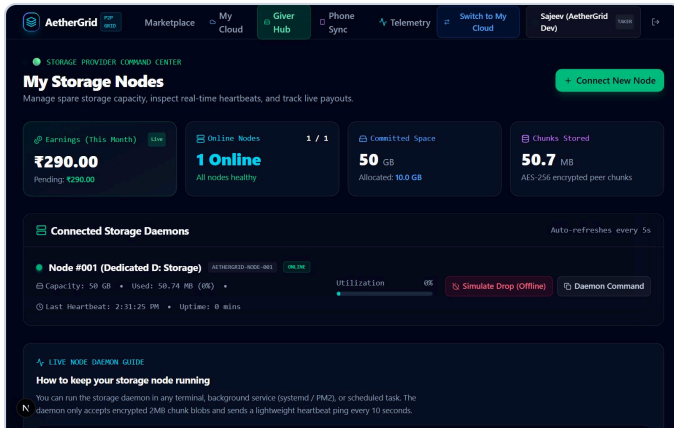
1. Landing Page & Peer Storage Marketplace



2. Taker Cloud Vault & File Manager

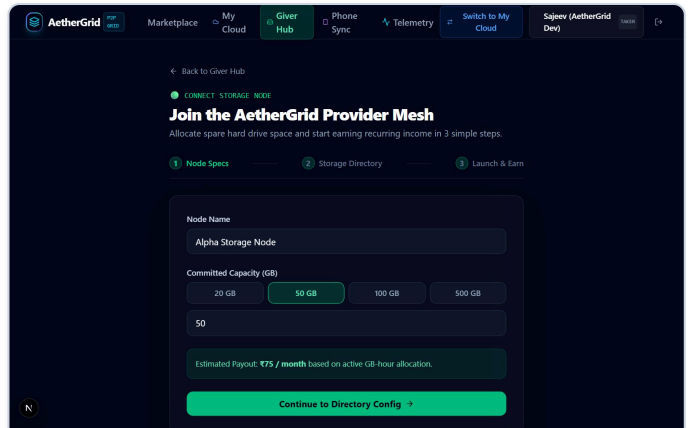
Public landing page showing P2P storage network economics, live capacity metrics, and zero-knowledge encryption guarantees.

Personal cloud storage console with 20GB tier allocation, drag-and-drop file upload, file status indicators, and one-click download.



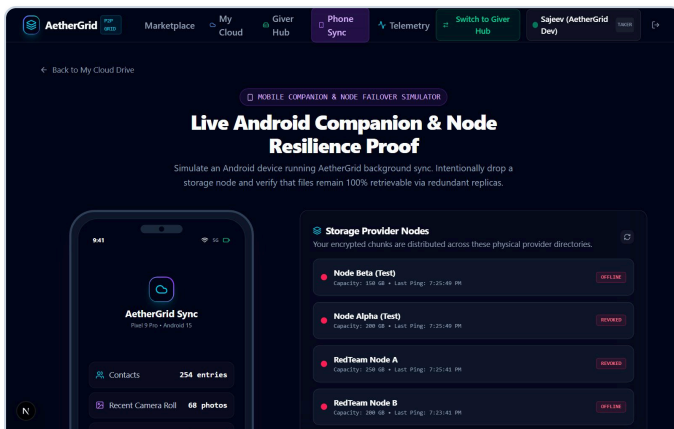
3. Giver Storage Node & Earnings Monitor

Live storage provider view monitoring physical disk Node #001 (D: drive), allocated GB-hours, real-time uptime, and monthly earnings in INR.



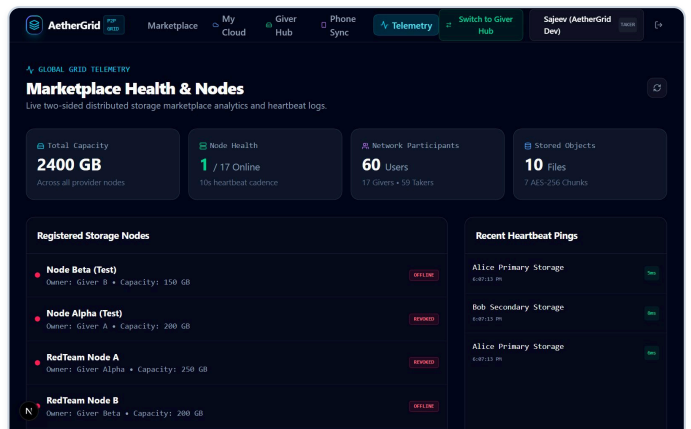
4. Giver Node Provisioning Wizard

Guided 3-step setup instructing providers how to register disk capacity and launch the local PowerShell node daemon.



5. Mobile Cloud Backup Simulator

Interactive phone simulator demonstrating background camera-roll backup, live sync progress, contact synchronization, and cloud vault recovery.



6. Admin Telemetry & Zero-Trust Monitor

Central control plane showing total users, active nodes, disk allocation ratios, zero-trust integrity status, and system audit logs.

3. Technical Architecture (How It's Built)

Front End

- **Next.js 15 App Router:** Server Components combined with interactive Client components for instant navigation.
- **React 19 & Tailwind CSS 4:** High-performance modern utility styling with bespoke dark-mode aesthetic.
- **Framer Motion & Lucide:** Micro-animations, progress meters, and dynamic telemetry cards.
- **Responsive UI:** Tailored views for mobile simulators, giver controls, and file explorers.

Back End

- **Node.js 24 + Web Crypto:** High-speed cryptographic primitives (HKDF, AES-256-GCM, SHA-256).
- **Storage Orchestrator:** Multi-replica 2MB chunking engine with automatic capacity reservation and disk safety limits.
- **Auth & Defense:** HttpOnly JWT sessions via jose, Bcrypt password hashing, and IP rate limiting.
- **60s Scoped Signed Tokens:** Single-use download

Database

- **Native SQLite (node:sqlite):** High-throughput zero-latency embedded database in WAL mode.
- **Relational Schema:** 10 tables managing users, sessions, storage nodes, chunks, files, allocations, and ledger payouts.
- **Dual-Platform Adaptability:** Runs against local NTFS physical disks (D:\AetherGridStorage) and Vercel serverless mounts (/tmp).

authorizations to prevent URL leakage.

4. What We'll Add Next (Next Sprint Roadmap)

- **Direct P2P WebRTC / gRPC Data Planes:** Allow takers and givers to transmit encrypted 2MB chunks directly between local daemons without proxying through the central web server, minimizing bandwidth costs and latency.
- **Native Mobile Background Backup Daemon:** Port the mobile simulator logic into a lightweight React Native / Swift / Kotlin mobile app with background camera-roll upload hooks and Wi-Fi-only battery-saving modes.
- **Client-Side Zero-Knowledge Encryption in WebAssembly:** Move the initial file encryption step entirely into the user's browser via WebCrypto / WASM, guaranteeing that unencrypted bytes never touch any network interface.
- **Automated Payment Gateway & Web3 Escrow:** Integrate Razorpay/Stripe autopay for monthly INR subscriptions alongside USDC micropayment smart contracts on Polygon for decentralized global provider payouts.